Internship Reports

---

1. General Information

Student Name: Moritz Albrecht Matriculation Number: 2134370

Internship Details

Organization: Bundesdruckerei GmbH

Supervisor: Matthias Matuschka

Working Hours: 20h/week
This corresponds to a part-

Remuneration: 1.304,00€/month

Main Tasks: Process
Management/Reporting, Process

Dates:Start: 01.11.2025 End: Language:

Crediting Details

25 ECTS (12 weeks full time)

Moritz Albrecht, 25.10.2026

Student Date

2. Report

2.1 Detailed Description

What did you do? Describe the organization and your tasks (2-3 pages):

2.2 Personal Reflection

What did you learn? Reflect on your learning experience, personal growth, realizations etc. (2-3 pages):

2.3 Weekly Reports

I. Process Management and Reporting

01 Analysing Customer Survey Results in *Excel*

02 Managing Business Opportunities through *JIRA*

03 Reporting on the Opportunity Pipeline

04 Adding Data Visualisation to Opportunity Reporting

II. Process Optimisation and Automation

05 Designing a Workflow for Automated *JIRA* Ticket Preparation

06 Starting a Python Script for JIRA Ticket Preparation

07 Automating Newsletter Archiving and Daily Digest Preparation

08 Evaluating *Assistent.iQ* for Research Support

III. Market Research

09 Researching the *Deutschland Stack* and Its Potential Applications

10 Researching Federal IT Consolidation

11 Comparing Online Identity Verification Services

12 Researching the *AI Act* and Its Implications for Digital Solutions

Appendix A Demonstrative pseudocode for Reports 05–06

Analysing Customer Survey Results in *Excel*

Period: 02.03.2026 to 15.03.2026

Task and context

I worked on the analysis of an anonymised customer survey to help assess how customers experienced the company's services and where improvements might be needed. The responses included standardised ratings, product information and written feedback. My task was to organise these different forms of information in *Excel* and prepare an evaluation that supported internal discussion.

Approach and contribution

I first examined the structure and completeness of the dataset, identifying the response categories available for each question and distinguishing unanswered questions from neutral responses. I then used *PivotTables* to count how many respondents selected each response category, from very dissatisfied to very satisfied, and calculated the corresponding percentages. For the written feedback, I developed thematic categories such as communication, price and product quality. I assigned comments to the relevant themes and counted how often each occurred, allowing a comment to cover more than one theme where appropriate. This made the feedback easier to compare while retaining the original wording for interpretation.

Since the responses were anonymized a separate task was to consider whether the available information suggested which customer had provided a response. I used product information and contextual clues in the written comments to infer possible customers. However, these clues generally did not establish their identity reliably, so the exercise largely depended on tentative guesses rather than confirmed assignments.

Outcome and skill development

The evaluation brought together quantitative patterns and qualitative feedback in a form that could support discussion of customer needs and possible improvements. It also made the basis and limitations of the findings more transparent.

This task applied statistical reasoning from my studies to a practical dataset. I developed my ability to interpret response distributions, handle missing information and assess how real world

subtleties, such as product differences, affected numerical comparisons. Analysing written responses also strengthened my ability to develop consistent categories while preserving differences in what respondents meant.

Attempting to identify customers highlighted the distinction between information recorded in a dataset and interpretations based on contextual clues. It showed why a plausible assignment should not be treated as a confirmed finding when the available evidence was limited.

Managing Business Opportunities through *JIRA*

Period: 17.11.2025 to 30.11.2025

Task and context

My daily work at Bundesdruckerei involved screening public procurement notices, identifying potential business opportunities and coordinating their assessment through the company's opportunity pipeline. The objective was to connect public-sector demand with the company's products, expertise and organisational responsibilities. This report focuses on a selected two-week period of these recurring responsibilities.

Approach and contribution

When reviewing a procurement notice, I assessed the requested solution and its relationship to Bundesdruckerei's portfolio. This required interpreting the administrative or technical problem behind the description. For example, a notice seeking secure identification for an online public service raised several distinct questions: did users need to prove their identity when registering, authenticate themselves on later visits, or sign documents electronically? Identity verification using the online ID function pointed towards *AusweisIDent*, offered by our sister cooperation D-Trust and Governikus; a requirement for electronic signatures pointed towards *sign-me*. Clarifying these functions helped me identify the relevant product area

I then assigned relevant opportunities to the appropriate key account management (KAM) team. The subsequent process involved several distinct assessments. KAM evaluated the economic potential of an opportunity; the responsible business unit assessed its fit with the portfolio and available resources; and the technical unit examined the requirements and initiated the development of a suitable solution where appropriate. My role was to support the correct routing of opportunities and maintain an overview of their progression.

I managed and monitored this process through the *JIRA* ticket tool. Tickets served as structured records linking the procurement information to its current status, responsible unit and subsequent evaluation. Monitoring the pipeline required following both individual opportunities and the handovers between teams. Accurate assignment and consistent documentation were important because later decisions depended on information gathered at earlier stages.

Outcome and skill development

My contribution supported a structured connection between market research and the company's commercial and technical evaluation processes. It made opportunities available to the relevant decision-makers and helped maintain traceability over their progress. This work developed my ability to distinguish commercial attractiveness from technical feasibility. It also strengthened my understanding of how a digital workflow represents an organisational process: categories, responsibilities and status transitions must correspond to the decisions people actually make.

Reporting on the Opportunity Pipeline

Period: 12.01.2026 to 25.01.2026

Task and context

Alongside the daily management of opportunities, I carried out the half-yearly analysis of the pipeline. The aim was to assess how successfully identified opportunities progressed through the organisation and to determine where delays occurred. This report examines one instance of that recurring reporting process.

Approach and contribution

I exported the relevant *JIRA* records into *Excel*. I first created an overview of opportunity volumes and outcomes to establish an initial picture of pipeline activity and success rates. I then examined the individual stages: identification of an opportunity, acceptance by the key account manager, assessment by the business and technical units, and preparation of an offer.

This more detailed analysis helped distinguish the volume of incoming opportunities from their progression through the evaluation process. A large number of identified opportunities did not necessarily imply a corresponding increase in offers. Examining the stages separately made it possible to investigate where and why opportunities were rejected, remained pending or advanced to the next step.

My analysis combined quantitative measures with qualitative information. I compared opportunity volumes, progression rates and the time spent at each stage. I also grouped the recorded reasons for rejection into recurring themes, such as portfolio fit, economic attractiveness or resource constraints. Counting these themes helped to show how frequently particular concerns were recorded, while retaining the original comments preserved the context. Together, these analyses identified stages where delays accumulated and helped explain why some opportunities did not progress.

Interpreting the results required care. Time spent in a given stage could include both active assessment and waiting, and cases still under review had to be distinguished from completed ones. In our team discussion, we considered incomplete status updates, handovers between units and competing priorities as possible explanations for delays. We also discussed whether the change of

federal government and uncertainty over public budgets might have influenced procurement timing or opportunity volumes.

Outcome and skill development

I presented the report within our team, where we discussed the findings and possible explanations in detail. My superiors also incorporated the analysis into stakeholder presentations. The results supported process optimisation by making delays and recurring rejection reasons easier to examine.

This task developed my skills in data preparation, aggregation and process analysis. I learned to move from an overall indicator to the underlying stages needed to explain it, and to distinguish an observed pattern from a causal explanation. The work demonstrated how operational data can inform both management communication and practical organisational improvements.

Adding Data Visualisation to Opportunity Reporting

Period: 01.06.2026 to 14.06.2026

Task and context

On my own initiative, I extended the opportunity reporting with visualisations that made developments over time and differences between pipeline stages easier to understand. The reporting already contained numerical information, but separate tables required viewers to reconstruct comparisons themselves. My aim was to make relevant patterns more accessible for analysis and stakeholder presentations.

Approach and contribution

I merged the available historical figures into one dataset, with consistent categories for reporting periods and pipeline stages. Drawing on my Information Visualisation lecture, I began with the questions viewers needed to answer: how had activity changed over time, how did the stages compare, and where did unusually long processing times occur? I treated visualisation as a way of communicating these relationships and guiding interpretation.

For historical developments, I used line charts with half-year reporting periods on the horizontal axis and opportunity counts on the vertical axis. Separate series represented selected pipeline stages. This made it easier to see whether changes in incoming opportunities were accompanied by similar changes in positively assessed cases or offers.

I also considered grouped bar charts for direct comparisons between stages and periods. Lines emphasised temporal development, while bars made individual quantities easier to compare. The choice depended on the question the presentation needed to answer.

The design applied the lecture's distinction between marks and visual channels. Position encoded time and quantity, while consistent colour distinguished the stages. I used clear axis labels, a shared scale and a limited number of series to reduce clutter. Direct labels helped viewers identify the lines without repeatedly consulting a separate legend.

The duration analysis also showed why averages needed context: one *JIRA* ticket that had remained unattended for almost a year strongly affected the reported figures. I therefore

considered box plots as a further extension, since they give a more nuanced picture of outliers and median values.

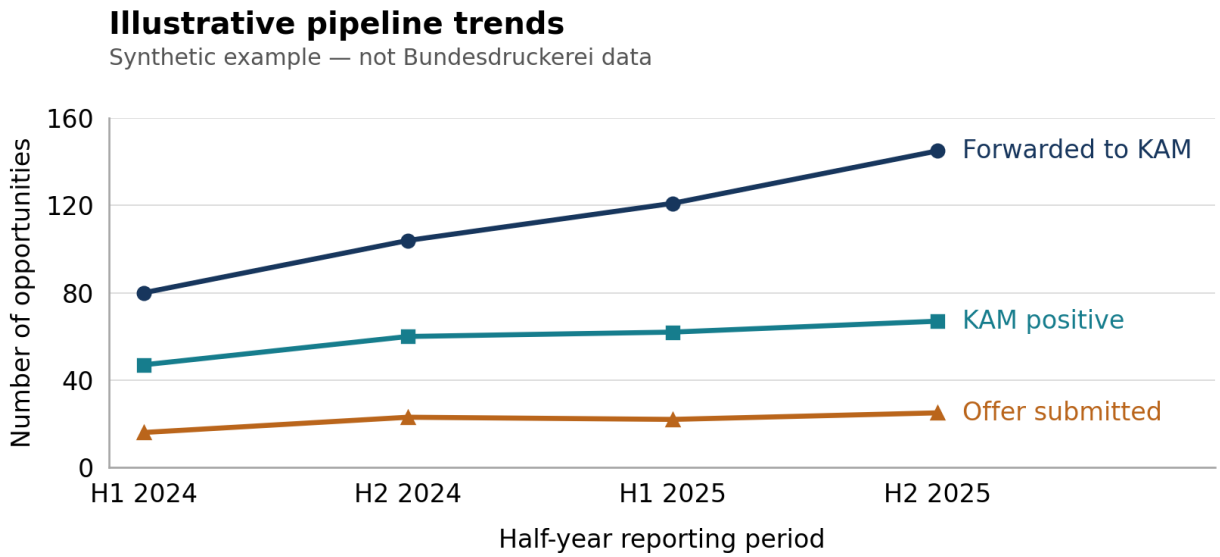


Figure 1. Example of the chart design using synthetic values. Counts refer to separate reporting periods, not a single cohort.

Outcome and skill development

My initiative added a historical and comparative perspective to the reporting and made its findings easier to communicate in presentations. It connected the analysis of the pipeline with the needs of the people interpreting the results.

The task directly applied knowledge from my Information Visualisation course. I developed my ability to select representations for specific analytical questions and to explain how design choices influence which patterns viewers notice.

Designing a Workflow for Automated JIRA Ticket Preparation

Period: 16.03.2026 to 29.03.2026

Task and context

Recording procurement opportunities in JIRA involved repeatedly transferring information from notices and assigning tickets to the responsible key account manager. After colleagues covered my task during an illness, I noticed differences in the entries. Alongside my recurring opportunity-management responsibilities, I investigated whether a small automation could make this preparation more consistent.

Approach and contribution

I focused on defining the workflow and its requirements. I limited the proposal to initial data entry from one recurring format of procurement PDF (Bekanntmachung). The target fields were client (Auftraggeber), a financial figure for expected revenue (Erwarteter Umsatz), award criteria (Zuschlagskriterien) and duration (Laufzeit).

I examined how the relevant passages corresponded to the JIRA fields. A notice could contain several monetary amounts and price weightings, so the financial figure needed to retain its meaning and source. A colleague would confirm whether it was suitable for expected revenue. Missing or ambiguous information would require manual clarification.

For assignment, I proposed using the existing Excel mapping between customers and responsible KAM contacts. An unambiguous customer match would supply the KAM name; missing or conflicting assignments would require review. The proposed script would prepare the entries for the JIRA form, while a colleague would check them, confirm the assignment, attach the PDF and submit the ticket.

I prepared a presentation for my superior explaining the inconsistent entries, the proposed processing steps and the points requiring human judgment. Subsequent commercial, business-unit and technical assessments would remain within the existing opportunity process.

Outcome and skill development

The workflow proposal developed my requirements-analysis skills: translating a workplace problem into inputs, processing rules and review conditions. Its intended benefit was less repeated copying and searching.

Starting a Python Script for JIRA Ticket Preparation

Period: 30.03.2026 to 12.04.2026

Task and context

Following the workflow design in Report 05, I started working on a Python script to prepare the initial JIRA ticket fields. This phase concerned translating the requirements into processing logic, alongside my recurring responsibilities, rather than redesigning the wider opportunity process.

Approach and contribution

I structured the work around three functions: reading the procurement notice, looking up the responsible KAM in Excel and preparing transfer to the JIRA form. For the extraction, I considered recurring headings and their associated text blocks within the selected document format. The logic needed to distinguish the buyer from other organisations and monetary figures from price weightings.

I specified checks for missing values and repeated required headings, which would stop processing for manual clarification. The Excel lookup would compare customer names while ignoring differences in case and extra whitespace. It would return a KAM only when the matching entries identified one responsible person without blank or conflicting assignments.

For transferring the information, I explored browser automation as a possible extension. The actual JIRA field selectors and assignment controls would need to be checked. A colleague would sign in normally, review the prepared values, confirm the financial figure and KAM selection, attach the PDF and submit manually.

Outcome and skill development

Implementation stalled because the required IT approval was not granted. I had started the Python work, but could not proceed to an approved implementation or test the connection to the live company JIRA system. The project therefore did not result in a deployed solution.

This phase developed my ability to decompose a workflow into functions and express validation rules explicitly. It also showed that technical progress depends on organisational approval. The pseudocode in Appendix A is included solely to demonstrate the proposed idea; it is not executable code or evidence of a completed implementation.

Automating Newsletter Archiving and Daily Digest Preparation

Period: 13.04.2026 to 26.04.2026

Task and context

Daily *Tagesspiegel Background* (TSB) newsletters provided relevant information for market research at Bundesdruckerei. Saving the editions and preparing them for internal circulation involved repeated manual work. I worked on automating these steps so that colleagues could access the newsletters through a consistent archive and a consolidated daily email.

Approach and contribution

I divided the process into two workflows: archiving each newsletter when it arrived and preparing a daily digest at a scheduled time. A forwarding rule could redistribute individual emails, but it would not provide the PDF archive and consolidated overview needed for subsequent research.

Using *Power Automate*, I configured a workflow that recognised incoming TSB newsletters through their sender and subject. It converted the email's HTML content into a PDF and saved it in a designated SharePoint folder. A naming convention combining publication date and topic, such as 2026-06-15_TSB_Digitalisierung_und_KI.pdf, made the archived editions easier to identify and retrieve.

The conversion required checking whether the resulting PDF preserved the newsletter's content and remained readable. I used the HTML version rather than plain text to retain more of the original structure and formatting, and checked sample PDFs against the source emails.

The second workflow collected the day's archived editions and assembled a consolidated email draft for review. Separating this scheduled step from the arrival-triggered archiving allowed newsletters received at different times to be included in one digest. I checked the draft for completeness before sending it to colleagues.

Outcome and skill development

The workflows reduced the manual effort involved in saving newsletters and preparing their daily circulation. The SharePoint archive provided a consistent location for subsequent research, while the consolidated email made the day's editions available together.

This task developed my ability to translate a recurring activity into explicit triggers, conditions and processing steps. It applied algorithmic thinking from my studies through the sequencing of actions and repeated processing of incoming newsletters.

Evaluating *Assistent.iQ* for Research Support

Period: 15.06.2026 to 28.06.2026

Task and context

I explored how Bundesdruckerei's internal AI assistant, *Assistent.iQ*, could support recurring research tasks. The objective was to improve the consistency of document summaries and preliminary assessments through structured instructions and examples. My contribution concerned developing and evaluating prompts for the assistant, rather than retraining its underlying model.

Approach and contribution

I began by selecting a narrowly defined task: producing a structured briefing from a supplied research document. The expected output included the main development, the organisations involved, relevant dates and potential business implications for Bundesdruckerei.

Initial attempts showed why a general instruction such as "summarise this document" was insufficient. An answer could be fluent while omitting an important qualification or presenting a possible implication as an established fact. I therefore developed a reusable prompt specifying the structure of the answer and requiring a distinction between source-supported information and interpretation.

I added examples demonstrating the desired output and how to handle missing information. This applied the distinction between instruction-only and few-shot prompting from my Intelligent User Interfaces (IUI) lecture on prompting-based interaction. If the document did not establish an implementation date or procurement status, the examples showed an explicit "not stated" response. I also requested references to relevant passages so that important claims could be checked.

To evaluate how well the prompts supported this task, I compared outputs against the original documents using a simple assessment table. The criteria included factual accuracy, coverage of relevant information, unsupported claims and usefulness for the intended reader. I revised the instructions in response to recurring errors and checked them on additional documents that had not guided the initial revisions.

Outcome and skill development

The work resulted in a reusable prompt template and an evaluation approach for AI-assisted research. It developed my ability to turn a broad expectation such as “a useful summary” into observable criteria and to revise instructions through systematic comparison. The task also made the IUI course’s emphasis on user control concrete: useful assistance depended on making outputs easy to verify and correct.

Researching the *Deutschland Stack* and Its Potential Applications

Period: 03.11.2025 to 16.11.2025

Task and context

I researched the *Deutschland Stack* as a proposed shared foundation for digital public services across federal, state and municipal administrations. A central aim was greater digital sovereignty: public institutions should retain control over their data and the systems on which their services depended. During this two-week period in November 2025, I examined the available plans, including the October 2025 federal *Modernisation Agenda*, and assessed where Bundesdruckerei's competencies could contribute.

Approach and contribution

I examined how the initiative connected common standards, reusable components and public-sector applications. Its scope was still contested: in a July 2025 *Tagesspiegel Background* commentary, Thilak Mahendran wrote, "Der Deutschland-Stack ist zur Projektionsfläche geworden." I therefore distinguished political objectives, the architecture governing how components interact, and the actual services proposed for implementation. The October 2025 federal *Modernisation Agenda* provided a more concrete basis for this assessment.

Planned components in October 2025

Selected elements of the federal *Modernisation Agenda*; this is not a final architecture or a list of completed deployments.

Component	Intended role
Digital identity and access	Digital identification, access management and *EUDI Wallet* integration.
Payments	Reusable payment functionality for administrative services.
Once-only data exchange	*NOOTS* to support reuse of administrative evidence and data.
Cloud and service marketplace	*DVC* infrastructure and a marketplace for shared services.
Planning and approvals	An AI-supported platform, with hydrogen infrastructure as a use case.

I related the *Stack* to service reuse (Nachnutzung) and the One-for-All (*Einer-für-Alle*; EfA) model used in implementing the *Online Access Act* (*Onlinezugangsgesetz*; *OZG*), under which one state or a group of states develops and operates a service for use by other administrations. I identified a potential coordination tension: shared *Stack* components could reduce duplication, while existing EfA services already had their own operating arrangements and interfaces.

As a concrete case, I examined the Hamburg-led platform for planning and approval processes for the hydrogen core network. The project had been announced in 2024, and an AI-supported planning and approval platform was included among the *Stack's* proposed components. I considered how document submission and administrative review might connect to shared identity, data-exchange and infrastructure services. These were possible integration paths; the plans did not establish that the platform was already fully integrated into the *Stack* during the research period.

These proposed shared services led to a longer discussion with my superiors about the *Stack's* goal of digital sovereignty. If an authority relied on a common identity or cloud service, could it retain control of its data and change suppliers without rebuilding the entire application? Open interfaces could make components easier to replace, while reusable software could reduce dependence on a single vendor. We also considered the practical limits: an authority still needed the expertise and contractual rights to operate, maintain or replace the technology. This gave the term "sovereignty" a concrete meaning in our assessment.

For Bundesdruckerei, I considered possible contributions in digital identities, secure access and trusted information exchange. I distinguished announced components from available services and concrete procurement opportunities when assessing the maturity of these possibilities.

Outcome and skill development

The research connected the *Stack's* objectives with planned components, a specific application and possible areas of company involvement. It provided a basis for monitoring responsibilities, interfaces and implementation decisions.

This work strengthened my ability to analyse the relationship between infrastructure, shared services and applications. I applied philosophical skills in clarifying digital sovereignty and distinguishing conceptual ambiguity from practical coordination problems. It also showed why technical proposals need to be assessed alongside existing institutional commitments.

Retrospective update

After the period covered by this report, the plans were further refined. On 18 March 2026, the IT Planning Council adopted a platform core covering identity and trust, data exchange, data retrieval, payments and a mailbox, alongside standards for *Stack* solutions.

Researching Federal IT Consolidation

Period: 01.12.2025 to 14.12.2025

Task and context

I researched the German federal IT consolidation programme to assess its implementation status, organisational responsibilities and potential relevance to Bundesdruckerei. The assignment involved understanding both the overall programme and individual digitalisation measures. The objective was to establish where administrative needs and ongoing developments might provide opportunities for the company.

Approach and contribution

I structured the research around the consolidation of IT services, the consolidation of IT operations and the bundling of procurement. This helped distinguish administrative applications, their supporting infrastructure and the arrangements for purchasing them. I also recorded the dates of the underlying documents, because the original programme structure did not necessarily reflect later institutional changes.

I documented individual measures in *Excel*, including their type, supporting sources, legal or administrative basis, responsible institutions and implementation status. Examples included electronic file management, federal payment services, procurement applications and shared infrastructure such as the *Bundescloud*.

An important challenge was establishing which information remained applicable. The creation of the Federal Ministry for Digitalisation and Government Modernisation (BMDS) in the new term meant that older references to ministries or departments needed to be checked against more recent responsibilities. I compared earlier programme documentation with newer information about the BMDS, the Federal Ministry of the Interior (BMI) and ITZBund, distinguishing a historical programme assignment from a current institutional responsibility.

I also distinguished between rollout, ongoing operation and further development. A programme being formally completed did not necessarily mean that implementation needs had disappeared. This distinction informed the assessment of possible Bundesdruckerei involvement: the relevant question was which concrete requirements remained and how they related to the company’s capabilities.

Outcome and skill development

The research resulted in a structured overview of consolidation measures, responsibilities and implementation questions. It provided a basis for identifying areas that warranted further investigation and for directing subsequent research towards the relevant institutions.

The task strengthened my ability to synthesise information from heterogeneous sources and assess its reliability and temporal relevance. It connected technical understanding of IT systems with institutional analysis. I applied conceptual and analytical skills from my studies to distinguishing programme objectives, technical functions and organisational responsibilities, and to explaining how these relationships affected the interpretation of a potential business opportunity.

Comparing Online Identity Verification Services

Period: 15.12.2025 to 04.01.2026

Task and context

I compared services for online identity verification in Germany, focusing on public-service portals. The reference product was *AusweisIDent*, developed and jointly offered by D-Trust, part of the Bundesdruckerei Group, and Governikus. I compared it with *WebID's eID* and *IDnow VideoIdent* from *IDnow GmbH* to examine which customer requirements favoured each approach.

Approach and contribution

I first defined the function being compared: confirming a person's identity remotely using an official identity document. This kept the comparison focused on identification during registration or an application. Routine account login and electronic signing were separate requirements; for example, D-Trust's *sign-me* could support a subsequent signature step, but would not replace the identification service.

I recorded the products in a comparison matrix covering the identification method, documents and equipment required, integration into an existing service and the steps the user had to complete. The main differences were as follows.

Service	Identification method	Practical requirement
*AusweisIDent Online*D-	Reads identity data from the eID chip;	Compatible eID document, PIN and reader
WebID eID	Reads the eID chip through the *My*	Compatible eID document, PIN and NFC
IDnow VideoIdent	A specialist checks the person and	Supported identity document, camera and

AusweisIDent offered an advantage over a video-based approach where users had a suitable eID document and PIN: identity data could be read from the chip without a live call. The service also avoided the need for the portal operator to obtain its own authorisation certificate. However, chip-based identification was not a unique selling point against *WebID's eID*, which used the same underlying online ID function. That comparison required closer attention to the app journey, integration effort, support and commercial terms.

For a portal serving users without an activated eID or available PIN, a video-based alternative could extend access, provided their documents were supported. This showed why the preferred solution depended on the intended users and the procedure's requirements. I could compare

documented functions, but public information did not support a reliable ranking of negotiated prices, completion rates or actual integration costs.

I summarised these trade-offs for internal discussion. The strongest case for *AusweisIDent* was as an eID component within an existing service whose users could use the online ID function. I treated that as a specific use case rather than a claim that it was superior to every competing identification method.

Outcome and skill development

The comparison clarified the relevant market and gave the team specific questions for assessing opportunities: which documents would users have, which identification method was required and how would the service fit into the customer's existing application?

This task combined economic reasoning about competing services with technical analysis of how they worked. I learned to distinguish a product's documented features from a customer advantage, and an advantage shared by several providers from a genuine differentiator.

Researching the *AI Act* and Its Implications for Digital Solutions

Period: 18.05.2026 to 31.05.2026

Task and context

I researched the European *AI Act* to assess its relevance to the development and use of AI-based solutions and to identify potential implications for Bundesdruckerei. The objective was to understand how regulatory requirements could affect public-sector applications and create demand for technical and organisational support.

Approach and contribution

My research centered on the questions that needed to be answered before assessing an AI application's relevance to Bundesdruckerei: What would the system be used for, which regulatory requirements would follow from that use, and who would be responsible for meeting them? I examined these questions using *Assistent.iQ*, Bundesdruckerei's document-based AI assistant for public authorities, as a concrete example. Drawing on the *AI Act* lectures in my Machine Ethics course, I considered how its regulatory assessment depended on the task an authority wanted it to perform. For example, using the assistant to summarise administrative guidance on public benefits would serve a different purpose from asking it to recommend whether a particular applicant should receive a benefit. I treated the latter as a hypothetical extension requiring a separate assessment of the intended use and potentially relevant high-risk requirements. This helped me examine the boundary between the product's document-assistance functions and possible customer requests for support with individual decisions.

I then considered how responsibilities would be divided between Bundesdruckerei as the product provider and a public-sector customer using the assistant. Continuing the public-benefits example, I considered a hypothetical administrative office using *Assistent.iQ* to summarise guidance when preparing replies to citizens. Bundesdruckerei would need to explain the product's intended functions and limitations, while the office would need to establish how staff should use and check its outputs. For example, the product's source references could help a caseworker compare an AI-generated summary with the underlying guidance before including it in a reply. The office would still need to decide who performed this check and who remained responsible for the final

response. This connected my research to a hypothetical customer need: introducing the product required both verifiable outputs and clear responsibilities within the customer's workflow.

Outcome and skill development

The research gave me a more precise basis for discussing AI applications and potential customer needs with the team. It connected regulatory requirements to questions about the intended workflow and the company's capabilities.

The task applied the contents of the Machine Ethics lectures by examining how the discussed concepts, like transparency, oversight and accountability, were required in practice. The main skill I developed was translating abstract regulatory and ethical concepts into concrete questions about product functionality and administrative workflows. In particular, I learned that providing source references supports verification, while effective oversight also requires clear responsibilities for checking and using the output.

Sketch of the proposed preparation workflow

The following pseudocode illustrates the idea described in Reports 05–06. It was prepared for explanatory purposes and does not reproduce a completed Python implementation. Form filling is shown as a conditional extension: live JIRA integration was not implemented or validated.

Document extraction and customer lookup

```
PROCEDURE PrepareTicket(procurementPDF, customerKAMTable)

    text ← ExtractText(procurementPDF)
    IF text cannot be read THEN
        STOP and request manual processing
    END IF

    sections ← IdentifyHeadingsAndTextBlocks(text)
    requiredSections ← FindBuyerFinancialCriteriaAndDuration(sections)

    IF a required section is missing or repeated THEN
        STOP and request manual clarification
    END IF

    Auftraggeber ← ExtractBuyerName(requiredSections)
    financialFigure ← ExtractAmountCurrencyAndSourceLabel(requiredSections)
    Zuschlagskriterien ← ExtractAwardCriteria(requiredSections)
    Laufzeit ← ExtractDuration(requiredSections)

    IF any required value is missing or ambiguous THEN
        STOP and request manual clarification
    END IF

    // A price weighting is not a monetary contract value.
    IF financialFigure represents a price weighting THEN
        STOP and request clarification of the financial figure
    END IF

    matches ← FindCustomerMatches(customerKAMTable, Auftraggeber,
                                  ignoring case and extra whitespace)

    IF matches do not identify exactly one responsible KAM
       OR any matching assignment is blank THEN
        STOP and request clarification of the KAM assignment
    END IF

    KAM ← responsible person identified by the mapping
```

Appendix A continued | Review and proposed transfer

```
    DISPLAY Auftraggeber, financialFigure and its source label,
            Zuschlagskriterien, Laufzeit and KAM

    colleague compares the information with the source documents
    colleague confirms or corrects the figure for Erwarteter Umsatz
    colleague confirms or corrects the KAM assignment

    IF required information remains unresolved THEN
        KEEP the information for manual clarification
        STOP
    END IF

    // Proposed extension, subject to IT approval and configuration.
    IF browser automation is approved and configured THEN
        OPEN Jira and WAIT for colleague to sign in and open the form
        ENTER the four reviewed values and the KAM name
        colleague reviews entries and confirms the KAM selection
        colleague attaches the source PDF
        colleague submits the ticket manually
    ELSE
        KEEP the reviewed information for manual transfer to Jira
    END IF

END PROCEDURE
```

The sketch deliberately omits library calls, selectors and other implementation details. Extracting a procurement amount does not establish expected company revenue; the colleague must confirm its suitability. The final ticket submission remains manual.